$$W = \frac{1}{2}k\left(h_2^2 - h_1^2\right) + mg\left(h_2 - h_1\right) . \qquad (7)$$

The first law of thermodynamics gives

$$\triangle Q = \triangle U + W . \qquad (8)$$

These equations and the given data yield

$$\triangle Q = \frac{k_1 k_2}{k_1 + k_2}H^2 + \frac{5}{4}mgH . \qquad (9)$$

4.

1. For voltage U between the sphere and shell, at radius r ($R_1 < r < R_2$) the electric-field magnitude is

$$= \frac{R_1 R_2}{R_2 - R_1}\ \frac{U}{r^2} . \qquad (1)$$

The field peaks at the surface, $r = R_1$. Write this field as $E(R_1)$; then

$$E(R_1) = \frac{R_2 U}{(R_2 - R_1)R_1} . \qquad (2)$$

By the given condition, $E(R_1)$ cannot exceed E_k, R_2 is given, (2) becomes

$$E_k = \frac{R_2 U}{(R_2 - R_1)R_1} \qquad (3)$$

or

$$U = E_k \frac{(R_2 - R_1)R_1}{R_2} . \qquad (4)$$

Choosing R_1 to maximize $(R_2 - R_1)R_1$ maximizes the withstand voltage U. Clearly, for

$$R_1 = \frac{R_2}{2} \qquad (5)$$

the voltage U reaches the maximum withstand value U_k. From (4), (5) we obtain

$$U_k = \frac{E_k R_2}{4} , \qquad (6)$$

Substituting the data gives

$$U_k = 155\ \text{kV} . \qquad (7)$$

Breakdown occurs when the AC peak equals U_k. The corresponding RMS voltage is

$$U_e = \frac{U_k}{\sqrt{2}}\ 110\ \text{kV} . \qquad (8)$$

2. The equivalent circuit is shown in the figure.